Streaming Data Architectures

Cloud Computing

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Contents

Learning Outcomes

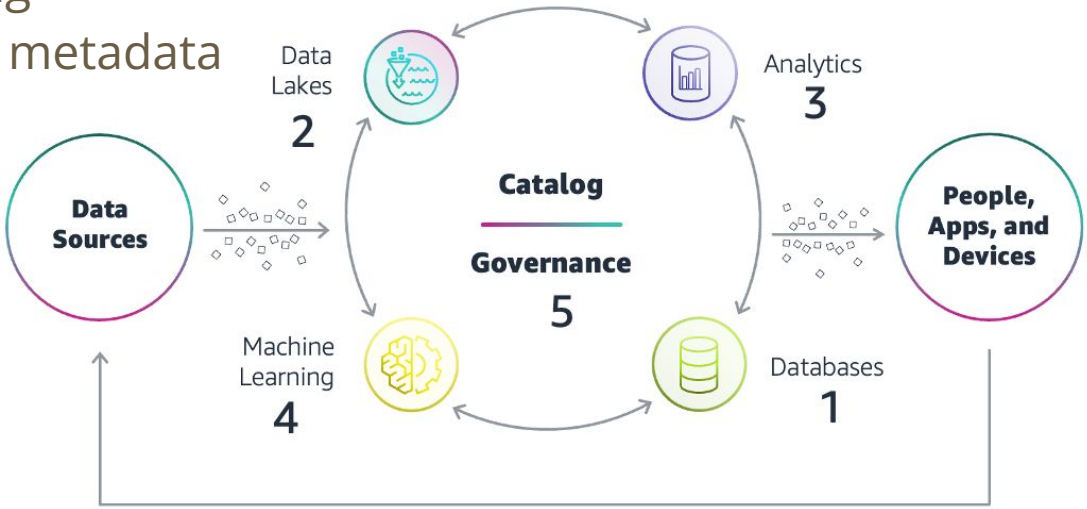
- Overview of Streaming Data Architectures
- Intro to Amazon managed streaming data services

Reading

- https://docs.aws.amazon.com/wellarchitected/latest/analytics-lens/analytics-lens.html
- https://docs.aws.amazon.com/whitepapers/latest/build-modern-data-streaming-analytics-architectures/build-modern-data-streaming-analytics-architectures.html
- https://docs.aws.amazon.com/msk/latest/developerguide/what-is-msk.html
- https://docs.aws.amazon.com/whitepapers/latest/build-modern-data-streaming-analytics-architectures/streaming-analytics-architecture-patterns-using-a-modern-data-architecture.html

Modern Data Architecture

- Purpose-built databases supporting a variety of data models
- Data lakes storing purpose-built data in native formats
- Comprehensive batch & real-time analytics
- Support for machine learning
- Central store of governance metadata



Key Characteristics

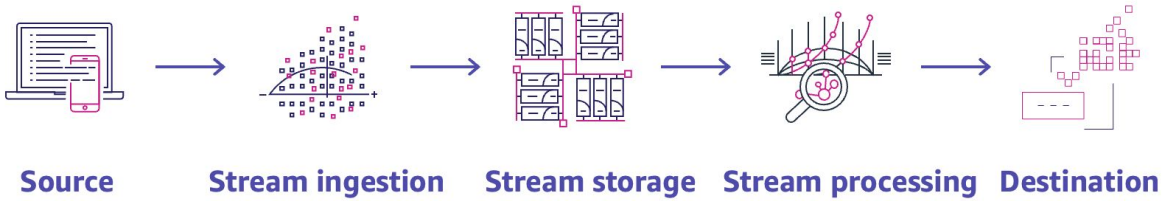
- **Scalable** throughput - storage infrastructure can adapt to changes in the rate of data flowing through the stream.
- **Dynamic** stream processor consumption & collaboration
- **Durable** - Real-time streaming systems should provide high availability and data durability.
- **Replay-able** - ability to independently read or replay records in the same order to stream processors and consumers
- **Fault-tolerance, checkpoint, and replay** - If the consuming application crashes, it can resume reading the stream from that point instead of having to start at the beginning.

Key Characteristics, cont.

- **Loosely coupled** - stream ingestion, stream producers, stream processors, and stream consumers can act independently of one another
- **Parallel processing** - multiple applications can consume the same stream concurrently
- **Security** - access granted on principle of least privilege. Data encrypted at rest & in transit

Intro to Data Streaming

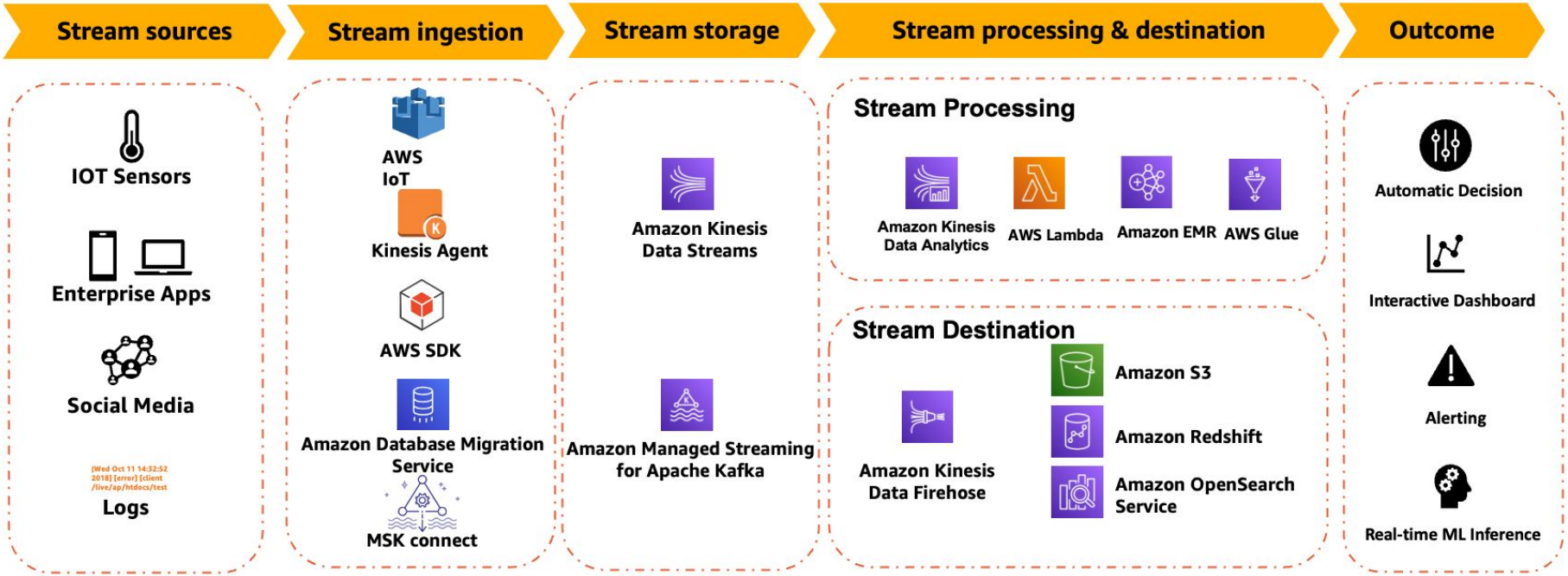
- Organizations often need to ingest, process, and analyze high volumes of high-velocity data from a variety of sources in real-time
- Data is stored in order of receipt and can be ‘replayed’
- Stream processing involves transforming data into a consumable state through data validation, cleanup, normalization, transformation, and enrichment.
- Modern data streaming architectures consist of 5 layers:



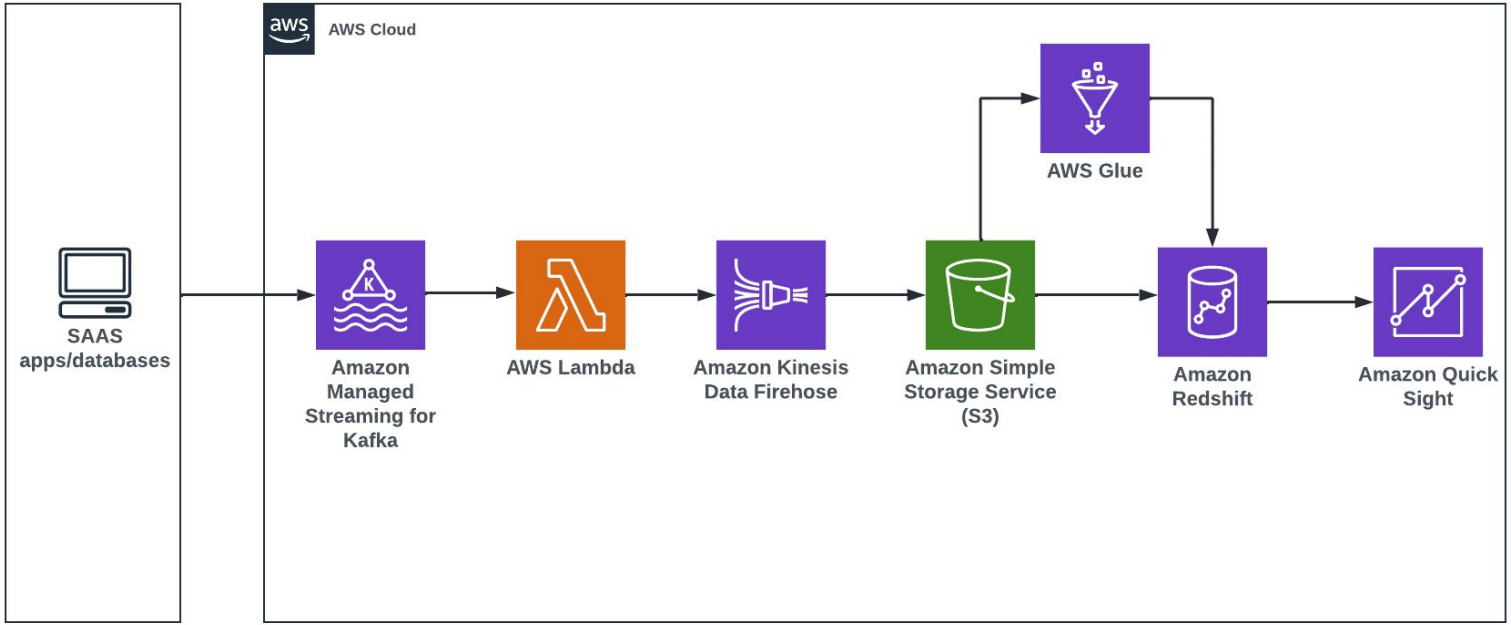
Use Cases

- Internet of Things (IoT) connected devices
  - Performance monitoring & failure detection
- Realtime data analytics & dashboards
- Data warehouses & data lakes
- Event-driven applications

AWS Streaming Data Lifecycle



Sample Real-time Analytics Architecture



https://docs.aws.amazon.com/whitepapers/latest/build-modern-data-streaming-analytics-architectures/streaming-analytics-architecture-patterns-using-a-modern-data-architecture.html

Custom streaming solutions

Engineers can develop and manage custom streaming data solutions on AWS EC2 & EMR, using open-source technologies such as Apache Kafka, Apache Spark, Apache Flume

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AWS Managed Streaming Data Services

AWS Kinesis offers a set of powerful, fully managed services that make it simple to load and analyze streaming data. It also enables you to build custom streaming data applications for specialized needs.

- **Amazon Kinesis Data Streams** - Scalable & durable service to collect & store data streams
- **Amazon Data Firehose** - Capture, transform, and load data streams into AWS data stores
- **Amazon Managed service for Apache Flink** - Process real-time data streams with SQL
- **Amazon Kinesis Video Streams** - Collect & store video streams
- **Amazon MSK** - Fully managed, highly available Apache Kafka service

Amazon Streaming Data Services

Easily collect, process, and analyze data streams in real time

Amazon Kinesis Data Streams

Collect and store data streams for analytics

Amazon Kinesis Data Firehose

Load data streams onto AWS data stores and third-party destinations

Amazon Kinesis Data Analytics

Analyze data streams with Amazon Kinesis Data Analytics Studio or Apache Flink

Amazon Kinesis Video Streams

Collect and store video streams for analytics

Amazon Managed Streaming for Kafka

Collect and store data streams for analytics

Amazon Kinesis Data Streams

- Fully managed service providing scalable & durable data stream ingestion
- Decouples data producers and consumers, allowing independent operations
- Provides reliable data delivery guarantees with configurable retention period
- Supports a variety of stream-processing frameworks
- Can receive streams of DB change data
- Can store streaming data directly to Amazon Redshift (data warehouse)
- Provides Kinesis Client Library (KCL) for consuming data and Kinesis Producer Library (KPL) for producing data

Kinesis Data Streams Terminology

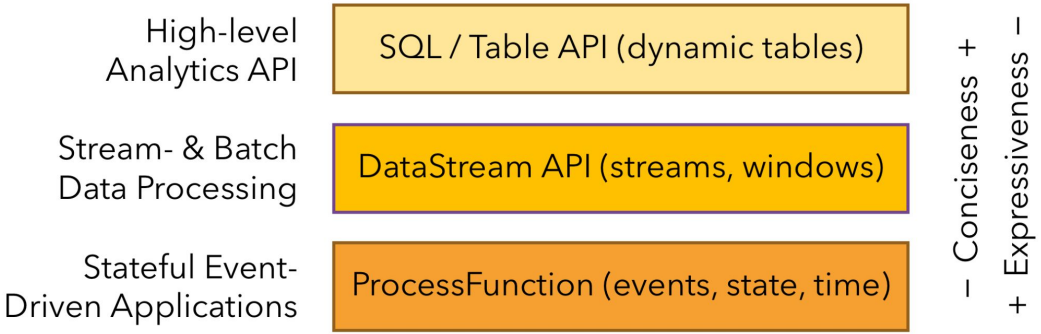
- **Producers** continually push data to Kinesis
- **Consumers** (e.g. Amazon Data Firehose) process the data in real time
- **Data stream** - a set of **shards**. Each shard is a sequence of data records, capable of handling 1 MBps of data ingestion.
- **Data record** - an **immutable** unit of data composed of a s**equence number**, a partition key, and a data blob
- **Partition key** - Value specified by producer and used to group data by shard within a stream
- **Capacity mode** - specifies how shards should be managed for throughput (**on-demand** or **provisioned**)
- **Retention period** - length of time that data records are accessible after being added to the stream

Amazon Data Firehouse

- Fully managed service to capture, transform, & load streaming data into storage destinations
- Can batch, compress, transform, & encrypt data streams
- Convert incoming data to standard-based formats, including columnar formats to optimize downstream analytics
- Supports dynamic, real-time partitioning of streaming data to optimize downstream analytics

Amazon Managed Service for Apache Flink

- Fully managed service to simplify implementation of Apache Flink applications
- Stateful, in-memory computations on data streams for real time analytics
- Rich support for time-related computations
- Layered APIs with different trade-offs between conciseness and expressiveness and targeting different use cases.



Amazon MSK

- Fully managed service simplifying applications that use Apache Kafka to process streaming data
- Abstracts complexity of provisioning, configuring, scaling, & maintaining Apache Kafka applications
- Can use native Apache Kafka APIs to integration with data sources and destinations
- Can serve as an event source, similar to AWS SQS or Kinesis